

Online Retail Retention Analysis – Project Workbook

Dataset: Online Retail II (UCI Machine Learning Repository) **Date Started:** April 12, 2026 **Analyst:** Joshua Paul Lasisi

Project Framing

What is the product?

An online retail platform serving both individual consumers and wholesale buyers across 43 countries. The platform sells physical goods — primarily home décor, gifts, and novelty items — through a transactional e-commerce model. Revenue is generated per purchase, not per subscription.

Who is the user?

Registered customers identified by Customer ID. Guest purchasers — rows without a Customer ID — are excluded because we cannot track their behavior over time. Without a Customer ID, there is no user, only a transaction. These rows are invisible to retention analysis.

What is the core value action?

Completing a purchase — represented by a unique Invoice with positive quantities. One invoice can contain multiple items (multiple StockCodes) but represents a single purchase session. A return (negative quantity) or cancellation (Invoice starting with "C") is the opposite of value delivery and is excluded from purchase counts.

What does the business want more of?

Repeat purchases from the same customer over time. Not just transaction volume — recurring value from the same user. A customer who buys once and never returns contributes less long-term value than one who buys four times a year at lower amounts.

What is this analysis trying to decide?

Whether and how quickly customers return after their first purchase — and what distinguishes customers who become repeat buyers from those who don't. The output should inform where to focus retention effort: immediately post-purchase, at a specific time window, or at a specific customer segment.

Dataset Observations

1. Dataset Time Span

Fact: The dataset spans 738 days — from December 2009 to December 2011.

Decision: We will build monthly cohorts across the full period, giving us approximately 24 cohort groups to analyze.

Why it matters: 24 cohorts is sufficient to detect both short-term patterns (does retention improve month over month?) and long-term trends (do early cohorts behave differently from later ones?). It also allows us to identify seasonal effects — if December cohorts consistently retain better than July cohorts, purchase timing may be influencing loyalty. Fewer than 12 cohorts would only reveal short-term patterns. 24 cohorts lets us distinguish temporary fluctuations from genuine retention trends.

2. Invoice-to-Customer Ratio

Fact: 53,628 unique invoices across 5,942 unique customers gives an average of approximately 9 invoices per customer over 2 years — roughly 4-5 purchases per customer per year.

Decision: We will not rely on this average. Instead we will build a purchase frequency distribution — showing how many customers made exactly 1, 2, 3, 4+ purchases — before drawing any conclusions.

Why it matters: This average almost certainly masks a skewed distribution. A small segment of high-frequency buyers — likely resellers or wholesale customers — probably inflates the mean significantly. If the majority of customers bought only once, the business has a severe retention problem that the 9-invoice average completely conceals. The distribution will tell us the truth. We state this as a hypothesis to be confirmed, not a conclusion already drawn.

3. Scale of the Dataset

Fact: 53,628 unique invoices across 5,942 unique customers and 5,305 unique products, spanning 43 countries.

Decision: We will use invoice volume per cohort per country as the primary activity measure, not product volume. Products cannot be reliably compared across 43 countries without complete description data. We will explore customer acquisition patterns per country to understand how the customer base is distributed — but direct country-to-country product comparisons will be flagged as unreliable without complete metadata.

Why it matters: The scale across 43 countries means local operational factors — shipping time, payment methods, return policies, customer service — may affect whether customers return, independent of product quality. Segmenting by country allows us to separate product-level retention problems from operational-level ones. Additionally, we cannot directly compare invoices with products across countries without knowing which products are most returned per location. The most sold and the most returned are different things, and conflating them would lead to wrong conclusions.

4. Negative Quantities — Returns and Cancellations

Fact: 229,950 rows have negative quantities, representing returns or cancellations. This is 21.5% of all rows — a significant proportion.

Decision: These rows will be excluded from purchase counts since they represent failed value delivery, not completed purchases. However, we will not discard this data. We will track return behavior separately — segmenting by cohort and by country — to identify whether certain time periods or geographies have disproportionately high return rates.

Why it matters: Returns are a behavioral signal, not just a data cleaning problem. Customers who return items frequently may be less likely to become repeat buyers — operational friction may be the hidden retention killer. If returns are growing faster than purchases over our 24-cohort window, that signals a product quality or operational problem that no amount of re-engagement effort will fix. Segmenting returns by cohort and country helps us narrow down whether the issue is with the product, the customer, or the operational experience in a specific location.

5. Missing Customer IDs and Descriptions

Fact: 242,000 rows — approximately 22.7% of all entries — are missing Customer IDs. Additionally, a smaller proportion of rows are missing product descriptions (approximately 0.4% of entries).

Decision: Rows missing Customer IDs will be excluded entirely from the retention analysis. The identity of each customer is the basis for all product-user relationship tracking. Without a Customer ID, the customer is a ghost — we cannot connect this transaction to any behavioral history. No advanced method can reliably recover identity from invoice data alone without risking false attribution.

For missing descriptions, we will attempt to impute using StockCode matching — if the same StockCode appears elsewhere in the dataset with a description, we will use that description. Any StockCode with no matching description anywhere in the dataset will be flagged. Since we are analyzing purchase behavior and not product content, description is metadata and not identity — dropping a whole record for a missing description is not appropriate. However, the volume of missing descriptions is noted as a data collection issue.

Why it matters: The identity of each customer is the basis for all product-user relationship analysis. Not having the Customer ID means we cannot track the product-user relationship. Losing 22.7% of rows is significant in absolute terms — it means our analysis reflects only the trackable customer base, not all buyers. This must be stated as a limitation in the final analysis. The missing descriptions point to a data entry automation gap, particularly significant given operations across 43 countries and nearly 5,700 unique products. This is an operational recommendation worth surfacing alongside the retention findings.

6. Geographic Spread

Fact: The dataset includes customers from 43 countries.

Decision: We will segment retention analysis by country where cohort sizes are large enough to be statistically meaningful. Countries with fewer than 50 customers will be grouped into an "Other" category to avoid drawing conclusions from too-small samples.

Why it matters: If retention rates vary significantly by country, it suggests that local operational factors — not just product quality — are affecting whether customers return. A country with high first-purchase volume but low repeat purchase rate may be experiencing shipping delays, payment friction, or a customer service gap specific to that region. This segmentation turns a geographic variable into an actionable diagnostic.

Metrics

Focus Metric

Definition: The percentage of activated customers who make a second purchase within 30 days of their first purchase.

What counts as valid:

- **Activation:** A customer who has completed at least one invoice with positive quantities and a valid Customer ID. Customers whose only transactions are returns or cancellations are not considered activated.
- **Second purchase:** A new invoice — different Invoice number from the first — containing at least one positive quantity line item. Cancellations (Invoices starting with "C") and returns (negative quantities) do not count.
- **Denominator:** All activated customers — every customer who made at least one valid purchase — regardless of whether they returned.

Why this metric: This metric directly shows us whether customers are returning, how fast they are returning, and how many of them are returning — all within the window most actionable for retention intervention. A 30-day window is appropriate for this product category: customers are unlikely to purchase daily, but 30 days is a realistic repeat purchase cycle for home goods and gifts. Daily or weekly windows would be too short; quarterly would obscure early churn signals.

Supporting Metrics

1. Percentage of customers who make a second purchase (at any time)

Definition: The percentage of activated customers who make at least one additional purchase at any point after their first — no time constraint.

Why it supports the focus: This metric gives us a larger view of the total repeat purchase potential. Since it is a superset of our focus metric, we can derive the focus metric directly from it. If the unconstrained repeat rate is 60% but the 30-day repeat rate is only 22%, that tells us customers do return — but slowly. The gap between these two metrics is itself an insight: it tells us whether the 30-day window is too tight or whether early re-engagement is genuinely missing.

2. Median time between first and second purchase

Definition: The median number of days between a customer's first valid invoice and their second valid invoice.

Why it supports the focus: This metric tells us where the natural repeat purchase window actually sits for this customer base — independent of our assumed 30-day window. We use median rather than mean because a small number of customers who wait 6-12 months before returning would

inflate the mean, making the typical customer appear slower to return than they actually are. If the median is 45 days, our 30-day focus metric may be measuring the wrong window. If the median is 18 days, 30 days is generous and we should be seeing higher conversion. This metric is a calibration check on the focus metric's time window.

3. Average purchases per customer per 30 days

Definition: Total valid invoices divided by total activated customers, calculated within each 30-day cohort window.

Why it supports the focus: This gives us a benchmark to judge the focus metric against across cohorts. The relationship between this metric and the focus metric should show direct variability over the 24 cohorts — if one rises, the other should rise. If average purchases per customer per 30 days is rising while the focus metric is flat, that signals that existing repeat buyers are buying more frequently — not that we are converting new repeat buyers. That tension is a distinct product problem and requires a different solution.

Guardrail Metric

Definition: The percentage of second-purchasers who go on to make a third purchase within 60 days of their second purchase.

Why: If we optimize aggressively for 30-day second purchases, we risk manufacturing repeat purchases through artificial means — discounts, urgency campaigns, or other short-term incentives — rather than through genuine value delivery. A customer who returns for a second purchase because of a discount did not necessarily find more value in the product. This guardrail catches that damage.

Customers who genuinely found value after their second purchase will return for a third. Customers who were pushed into a second purchase artificially will not. If our focus metric rises while this guardrail drops, we are not building retention — we are borrowing it from the future.

What it protects: Genuine customer loyalty — not just repeat transaction volume. The goal is not to get customers to buy twice. It is to get customers to become habitual buyers. The third purchase is the clearest available signal that habit has formed.

Why this is measurable from our data: We have Invoice, Customer ID, and InvoiceDate for all valid transactions. We can calculate each customer's first, second, and third purchase dates directly from these columns. This metric requires no external data.

What Comes Next

The following hypotheses will be tested in the analysis, in this order:

1. Most customers do not return after their first purchase — confirming whether this is an acquisition-heavy platform with low compounding value.
2. Customers who return, return quickly — informing whether retention efforts should begin immediately post-purchase.
3. The largest drop-off happens immediately after the first purchase — identifying whether the purchase experience itself is the primary bottleneck.
4. One-time buyers behave differently from repeat buyers in their first session — enabling persona-based targeting.
5. Second-purchasers are significantly more likely to keep buying than first-purchasers — confirming whether the second purchase is the critical conversion point.
6. Returns and cancellations correlate with lower retention — testing whether operational friction is a hidden retention killer.
7. Retention varies significantly by country — identifying whether local operational factors affect repeat purchase behavior.

Each hypothesis will be tested with a specific query, a confirmation condition, a rejection condition, and a recommended action if confirmed.